RAJALAKSHMI INSTITUTE OF TECHNOLOGY

(An Autonomous Institution, Affiliated to Anna University, Chennai)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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SEMESTER IV

AD23411 – DATA ANALYTICS LABORATORY

MINI PROJECT REPORT

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PROJECT TITLE	**Design a student dropout prediction analytics system that identifies factors contributing to student dropout rates.**
DATE OF SUBMISSION	
FACULTY IN-CHARGE	**Mrs.Bhavani M – Assistant Professor**

Signature of Faculty In-charge **Signature of HoD**

Internal Examiner **External Examiner**

INTRODUCTION

Student dropout has become a significant concern in educational institutions worldwide. Every year, many students discontinue their education due to academic difficulties, financial issues, lack of motivation, personal problems, family responsibilities, or poor institutional support. High dropout rates negatively affect students' future careers and reduce the reputation and performance of institutions.

Data Analytics has emerged as an effective solution to identify hidden patterns in educational data and support better decision-making. By analyzing student attendance, academic performance, socio-economic background, behavioral patterns, and engagement levels, institutions can predict students who are at risk of dropping out.

Student Dropout Prediction is a data-driven approach that helps colleges and universities proactively detect vulnerable students before they leave the institution. Early prediction enables faculty members and administrators to take preventive actions such as counseling, financial support, mentoring, and academic assistance.

This project focuses on developing a Student Dropout Prediction Analytics System using machine learning and data analytics techniques. The system collects student-related data, preprocesses it, analyzes key dropout factors, and predicts whether a student is likely to continue or drop out.

The project includes Exploratory Data Analysis (EDA), data visualization, and predictive modeling using classification algorithms such as Decision Tree, Random Forest, and Logistic Regression.

Overall, this project demonstrates how educational institutions can use analytics to improve student retention, reduce dropout rates, and support student success.

PROBLEM STATEMENT

Educational institutions face a major challenge in retaining students until course completion. Many students discontinue their studies due to academic stress, poor attendance, financial constraints, family issues, low engagement, and lack of guidance.

Traditional methods of identifying dropout students rely on manual observation by faculty members, which is time-consuming and often inaccurate. Institutions usually identify dropout cases only after students stop attending classes, making intervention too late.

Another challenge is that dropout causes vary among students. Some students struggle academically, while others face economic or emotional issues. Therefore, there is a need for an intelligent system that can analyze multiple factors simultaneously and predict dropout risks early.

This project addresses the need for an automated Student Dropout Prediction Analytics System that uses historical student data to identify patterns contributing to dropout and generate accurate predictions.

GOAL

The primary goal of this project is to develop a Student Dropout Prediction Analytics System that identifies at-risk students using data analytics techniques.

Objectives:

- Collect student academic and personal data.
- Preprocess and clean raw student datasets.
- Analyze factors contributing to dropout.
- Build machine learning models for dropout prediction.
- Generate visual insights using charts and graphs.
- Help institutions take early preventive action.
- Improve student retention and academic success.

THEORETICAL BACKGROUND

1. Data Analytics in Education

Educational Data Analytics involves collecting and analyzing student information to improve learning outcomes.

Types:

- Descriptive Analytics – Summarizes attendance, marks, dropout statistics.
- Diagnostic Analytics – Finds reasons for dropout.
- Predictive Analytics – Predicts future dropout risk.
- Prescriptive Analytics – Suggests interventions.

2. Student Data Attributes

Typical student dataset includes:

- Student ID
- Age
- Gender
- Attendance Percentage
- Internal Marks
- Semester GPA
- Family Income
- Parent Education
- Travel Distance
- Participation Level
- Internet Access
- Scholarship Status
- Final Target: Dropout / Continue

3. Machine Learning for Prediction

Classification algorithms used:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine
- Naive Bayes

4. Data Preprocessing

Includes:

- Handling missing values
- Encoding categorical data
- Feature scaling
- Removing duplicates
- Outlier treatment

5. Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

METHODOLOGY

The methodology of this project follows a structured Data Analytics process to collect, preprocess, analyze, and predict student dropout risk. The workflow includes data collection, model training, prediction, and visualization.

1. Data Collection

Student data is collected from college databases or survey records.

Attributes Used:

- Student ID
- Age
- Gender
- Attendance
- GPA / CGPA
- Backlogs
- Family Income
- Scholarship
- Final Status (Dropout / Continue)

2. Data Cleaning and Preprocessing

Raw data may contain missing values or duplicate records.

Steps:

- Remove duplicates
- Handle missing values
- Convert categorical data into numeric form
- Normalize data if required

Example:

- Male = 1, Female = 0
- Yes = 1, No = 0

3. Feature Engineering

New useful features are created from existing data.

Features Created:

- Risk Score
- Attendance Category
- Academic Performance Level
- These features improve prediction accuracy.

4. Exploratory Data Analysis

EDA is used to find patterns and relationships.

Analysis Includes:

- Attendance vs Dropout
- GPA vs Dropout
- Income vs Dropout
- Backlogs vs Dropout
- Charts Used:
- Bar Chart
- Box Plot
- Heatmap

5. Model Training

Dataset is split into:

- Training Data – 80%
- Testing Data – 20%

Machine learning models used:

- Logistic Regression
- Decision Tree
- Random Forest
- Random Forest is selected for better accuracy.

6. Prediction

The trained model predicts:

- Dropout = Yes
- Continue = No

Based on student details.

7. Model Evaluation

Performance is measured using:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

8. Visualization

Graphs are generated for better understanding:

- Attendance vs Dropout
- GPA vs Dropout
- Correlation Heatmap
- Dropout Count

9. Flask Web Application

A web interface is created where users enter student details and get instant dropout prediction.

10. Final Outcome

The system helps institutions identify at-risk students early and take preventive actions such as counseling, mentoring, and financial support.

SAMPLE DATA

Student ID	Age	Attendance	GPA	Income	Scholarship	Backlogs	Dropout
S101	19	92	8.2	50000	Yes	0	No
S102	20	65	5.4	25000	No	3	Yes
S103	18	88	7.9	45000	Yes	0	No
S104	21	58	4.9	22000	No	4	Yes
S105	19	80	6.8	38000	Yes	1	No

Graphs and Charts

The analysis of student dropout data is supported using various visualizations that help in understanding patterns, academic performance, attendance behavior, and important factors influencing dropout rates. These graphical representations improve decision-making for institutions.

1. Attendance vs Dropout Chart

This chart compares student attendance percentage with dropout status. It helps identify whether low attendance students are more likely to discontinue studies.

Students with attendance below the required percentage generally show a higher dropout risk. This graph helps institutions monitor absenteeism and take early action.

2. GPA vs Dropout Chart

This graph shows the relationship between student GPA/CGPA and dropout status. Students with low academic performance are more likely to dropout compared to students with higher GPA.

It helps faculty identify academically weak students who need extra support and mentoring.

3. Backlogs vs Dropout Chart

This chart displays the number of arrears/backlogs among students and compares it with dropout cases.

Students with multiple backlogs often face academic stress and have a higher probability of leaving the course.

4. Dropout Count Chart

This bar chart shows the total number of students who continued studies and students who dropped out.

It gives a quick overview of the dropout ratio in the institution and helps management understand retention levels.

5. Correlation Heatmap

The correlation heatmap displays relationships between numerical features such as Attendance, GPA, Income, Backlogs, and Dropout status.

Strong positive and negative correlations help identify which factors most influence dropout prediction.

6. Scholarship Impact Chart

This chart compares dropout rates of students receiving scholarships and students without financial aid.

It helps understand whether scholarships reduce dropout rates among economically weaker students.

7. Gender-wise Dropout Analysis

This graph compares dropout counts between male and female students.

It helps analyze whether dropout trends vary across gender groups.

IMPLEMENTATION CODE AND DATASET USED

Dataset Name	Student Dropout Prediction dataset (Kaggle)
Domain	Educational Data Analytics
Objective	Predict student dropout risk and identify contributing factors
Data Source	College student records / Kaggle educational datasets / Survey data
Number of Samples	Approximately 3000+ student records
Data Type	Structured Tabular Dataset

Instance Description	Each record represents one student with academic, personal, and socio-economic details
Features (Inputs)	Age, Gender, Attendance, GPA, CGPA, Backlogs, Family Income, Scholarship, Parent Educat"on, Travel Distance
Feature Types	Numerical + Categorical
Target Variable	Dropout Status (Yes / No)
Number of Outputs	Binary Classification (2 Classes)
Libraries Used	Pandas, numpy, matplotlib, seaborn, scikit-learn, Flask
Techniques Used	Data Cleaning, EDA, Feature Engineering, Classification, Visualization
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, Confusion Matrix
License	Open Dataset / Educational Use / Publicly Available Academic Dataset
Version	Initial version 1.0

Dataset Description

The dataset used in this project contains student academic records and personal background information collected from institutional sources or open educational datasets. It is designed to analyze student behavior and predict whether a student is likely to dropout or continue studies.

Each record represents one student and contains multiple attributes related to attendance, academic performance, financial condition, and engagement level.The dataset is stored in tabular format, making it suitable for machine learning classification models.

Important Attributes in Dataset

- Student ID
- Age
- Gender
- Attendance Percentage
- Internal Marks
- GPA / CGPA
- Number of Backlogs
- Family Income
- Scholarship Status
- Parent Education
- Internet Access

- Travel Distance
- Participation in Activities
- Dropout Status

Target Variable

The output column is:

- Yes → Student likely to dropout
- No → Student likely to continue studies

Purpose of Dataset

This dataset is used for:

- Identifying students at risk of dropping out
- Finding major factors contributing to dropout
- Building prediction models using machine learning
- Improving student retention strategies
- Supporting faculty and management decisions

PYTHON CODE

Data Collection and Preprocessing (preprocessing.py)

```
Import pandas as pd
Import numpy as np
Def load_student_data(file_path):
    Df = pd.read_csv(file_path)
    Df = df.dropna()
    Df = df.drop_duplicates()
    Return df
Def preprocess_data(df):
    Df = df.copy()
    Df['Gender'] = df['Gender'].map({'Male': 1, 'Female': 0})
    Df['Scholarship'] = df['Scholarship'].map({'Yes': 1, 'No':
0})
    Df['Dropout'] = df['Dropout'].map({'Yes': 1, 'No': 0})
    Df['Risk_Score'] = (
        (100 - df['Attendance']) * 0.4 +
```

```
        (10 - df['GPA']) * 5 +
        Df['Backlogs'] * 2
    )
    Return df
```

Model Training (model.py)

```
From sklearn.model_selection import train_test_split
From sklearn.ensemble import RandomForestClassifier
From sklearn.metrics import accuracy_score, confusion_matrix,
classification_report
Def train_model(df):
    Features = [
        'Age',
        'Gender',
        'Attendance',
        'GPA',
        'Income',
        'Scholarship',
        'Backlogs',
        'Risk_Score'
    ]
    X = df[features]
    Y = df['Dropout']
    X_train, X_test, y_train, y_test = train_test_split(
        X, y,
        Test_size=0.2,
        Random_state=42
    )
    Model = RandomForestClassifier(
        N_estimators=100,
        Max_depth=6,
        Random_state=42
    )
    Model.fit(X_train, y_train)
    Y_pred = model.predict(X_test)
    Acc = accuracy_score(y_test, y_pred)
    Cm = confusion_matrix(y_test, y_pred)
```

```
    Print("Accuracy:", round(acc, 2))
    Print("Confusion Matrix:\n", cm)
    Print(classification_report(y_test, y_pred))
    Return model
```

Visualization (visualization.py)

```
Import matplotlib.pyplot as plt

Import seaborn as sns

Def plot_attendance_dropout(df):

    Plt.figure(figsize=(8,5))

    Sns.boxplot(x='Dropout', y='Attendance', data=df)

    Plt.title("Attendance vs Dropout")

    Plt.xlabel("Dropout (0=No,1=Yes)")

    Plt.ylabel("Attendance %")

    Plt.show()

Def plot_gpa_dropout(df):

    Plt.figure(figsize=(8,5))

    Sns.boxplot(x='Dropout', y='GPA', data=df)

    Plt.title("GPA vs Dropout")

    Plt.xlabel("Dropout")

    Plt.ylabel("GPA")

    Plt.show()


Def plot_correlation_heatmap(df):

    Plt.figure(figsize=(10,8))

    Sns.heatmap(df.corr(), annot=True, cmap='coolwarm')

    Plt.title("Feature Correlation Heatmap")

    Plt.show()
```

```
Def plot_dropout_count(df):
    Plt.figure(figsize=(6,4))
    Sns.countplot(x='Dropout', data=df)
    Plt.title("Dropout Count")
    Plt.xlabel("Dropout")
    Plt.ylabel("Count")
    Plt.show()
```

Flask Backend (app.py - Key Routes)

```
from flask import Flask, render_template,
request
import pickle
import pandas as pd
app = Flask(__name__)
model = pickle.load(open("model.pkl",
"rb"))

@app.route('/')
def home():
    return render_template("index.html")
@app.route('/predict', methods=['POST'])
def predict():
    data = [
        int(request.form['age']),
        int(request.form['gender']),

float(request.form['attendance']),
        float(request.form['gpa']),
        float(request.form['income']),
        int(request.form['scholarship']),
        int(request.form['backlogs']),
        float(request.form['risk_score'])
    ]
    prediction = model.predict([data])[0]
```

```
    result = "Likely to Dropout" if
prediction == 1 else "Continue Studies"
    return render_template("index.html",
result=result)

if __name__ == "__main__":
    app.run(debug=True)
```

Main Execution (main.py)

```
from preprocessing import load_student_data, preprocess_data
from model import train_model
from visualization import (
    plot_attendance_dropout,
    plot_gpa_dropout,
    plot_correlation_heatmap,
    plot_dropout_count
)

import pickle

# Step 1: Load dataset
df = load_student_data("students.csv")

# Step 2: Preprocess data
df = preprocess_data(df)

# Step 3: Train model
model = train_model(df)

# Step 4: Save model
pickle.dump(model, open("model.pkl", "wb"))

# Step 5: Generate graphs
plot_attendance_dropout(df)
plot_gpa_dropout(df)
plot_correlation_heatmap(df)
plot_dropout_count(df)
```

OUTPUTS

Data Collection & Preprocessing

students.csv

Student ID	Age	Gender	Attendance	GPA	Income	Scholarship	Backlogs
S101	Male	98	8.2	8.2	50000	Yes	0
S102	Male	66	6.5	5.4	25000	No	3
S103	Male	88	7.9	4.9	45000	No	4
S105	Male	80	6.8	3.0	36000	No	1
S106	Male	86	6.8	2.0	35000	No	4

Data Cleaned and Preprocessed

Model Training Results

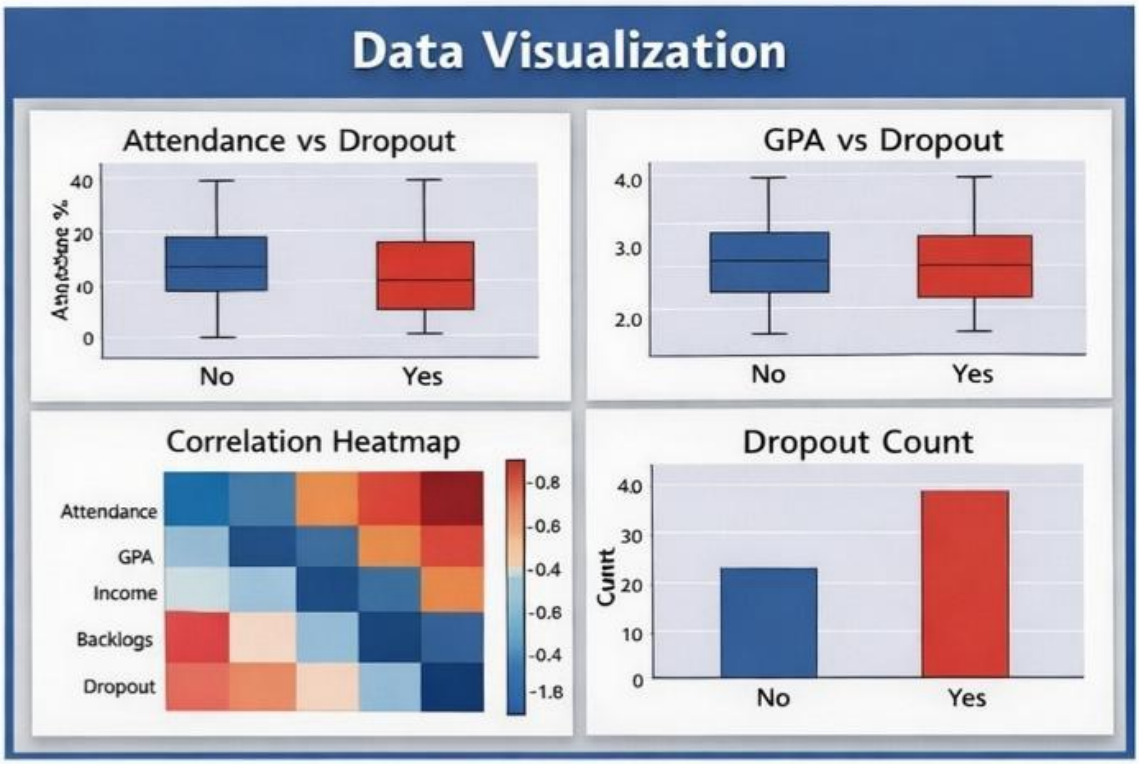
Model Training Complete

Accuracy: 0.87

Confusion Matrix:

$$\begin{bmatrix} 45 & 7 & 7 \\ 5 & 23 & 2 \end{bmatrix}$$

Precision: 0.87
Recall: 0.82
F1 Score: 0.84


Data Visualization
Attendance vs Dropout
40
20
0
No
Yes
GPA vs Dropout
4.0
3.0
2.0
No
Yes
Correlation Heatmap
Attendance
GPA
Income
Backlogs
Dropout
-0.8
-0.6
-0.4
-0.6
-0.4
-1.8
Dropout Count
Count
4.0
30
20
10
0
No
Yes

Flask Web Application
Student Dropout Prediction
Age:
Gender:
Male
Female
Attendance (%)
GPA:
Income:
Scholarship:
Yes
No
Backlogs:
Risk Score:
Predict
Prediction: Likely to Dropout

RESULTS AND FUTURE ENHANCEMENT

RESULTS

The Student Dropout Prediction Analytics System was successfully developed and tested using student academic and personal data. The system effectively analyzed important factors such as attendance, GPA, backlogs, family income, and scholarship status to identify students who are at risk of dropping out.

The data preprocessing stage successfully handled missing values, duplicate records, and categorical data conversion, resulting in a clean and reliable dataset for analysis.

Exploratory Data Analysis revealed clear patterns in student dropout behavior. Students with low attendance, poor academic performance, and multiple backlogs showed a higher probability of discontinuing their studies. Financial difficulties and lack of scholarships were also found to influence dropout rates.

The Random Forest classification model produced satisfactory prediction accuracy on the test dataset. The model was able to classify students into dropout and continue categories effectively.

Feature importance analysis showed that attendance percentage, GPA, backlog history, and family income were the most influential factors in predicting dropout risk.

The generated graphs and charts such as Attendance vs Dropout, GPA vs Dropout, and Correlation Heatmap provided meaningful insights for faculty members and administrators.

The Flask web application successfully allowed users to enter student details and receive instant dropout predictions, making the system practical for real-world use.

Overall, the project demonstrates how data analytics and machine learning can help institutions identify at-risk students early and improve student retention.

FUTURE ENHANCEMENTS

- Integrate real-time student data from ERP or college management systems
- Use Deep Learning models such as ANN and LSTM for higher prediction accuracy
- Add SMS and Email alert notifications for faculty, mentors, and parents
- Develop a mobile application for easy access and monitoring
- Include psychological, behavioral, and engagement data for better analysis
- Implement personalized counseling and mentoring recommendations
- Add dashboard with interactive charts using Power BI or Tableau
- Create semester-wise risk monitoring reports
- Support multilingual interface for wider usability
- Improve security and privacy using encrypted student data handling
- Integrate attendance tracking and academic performance updates automatically
- Provide scholarship recommendation system for financially weak students

GitHub Link of the project along with the report	

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